Completing the Square & Equations (Multiple Choice)

1. Rewrite in vertex form:
$$y = x^2 + 6x + 2$$
   (a) $y = (x+3)^2 + 7$
   (b) $y = (x-3)^2 - 7$
   (c) $y = (x+3)^2 - 7$
   (d) $y = (x+6)^2 - 34$

2. Solve by completing the square:
$$x^2 - 4x + 1 = 0$$
   (a) $x = 2 \pm \sqrt{5}$
   (b) $x = -2 \pm \sqrt{5}$
   (c) $x = 4 \pm \sqrt{3}$
   (d) $x = 2 \pm 3$

3. Rewrite in vertex form:
$$y = 2x^2 + 8x + 3$$
   (a) $y = 2(x+4)^2 - 29$
   (b) $y = 2(x-2)^2 - 5$
   (c) $y = 2(x+2)^2 + 3$
   (d) $y = 2(x+2)^2 - 5$

4. Solve:
$$x^2 + 10x + 1 = 0$$
   (a) $x = -5 \pm \sqrt{24}$
   (b) $x = 5 \pm \sqrt{24}$
   (c) $x = -10 \pm 1$
   (d) $x = 5 \pm 24$

5. Rewrite in vertex form:
$$y = 3x^2 - 12x + 7$$
   (a) $y = 3(x+2)^2 - 5$
   (b) $y = 3(x-2)^2 + 7$
   (c) $y = 3(x-2)^2 - 5$
   (d) $y = 3(x+2)^2 + 7$

6. Solve by completing the square:

$$x^2 + 8x + 20 = 0$$

(a) $x = -4 \pm 2$

(b) $x = -4 \pm 2i$

(c) $x = 4 \pm 2i$

(d) $x = -8 \pm 4$

7. Rewrite:

$$y = -x^2 + 6x - 2$$

(a) $y = -(x + 3)^2 + 7$

(b) $y = -(x - 3)^2 + 7$

(c) $y = -(x - 3)^2 - 11$

(d) $y = -(x + 6)^2 + 34$

8. Solve:

$$2x^2 + 4x - 3 = 0$$

(a) $x = -1 \pm \sqrt{2}$

(b) $x = 1 \pm \sqrt{2}$

(c) $x = -1 \pm \frac{\sqrt{10}}{2}$

(d) $x = 1 \pm \frac{\sqrt{10}}{2}$

Answer Key

1. C
2. A
3. D
4. A
5. C
6. B
7. B
8. C